

Content Today:-

- 1) Rain Water Trapping
- 2) first Missing positive

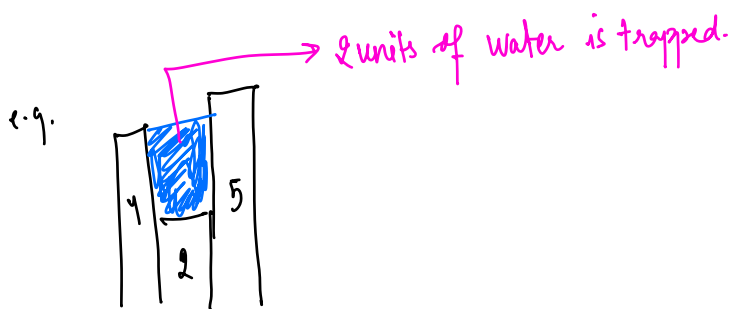
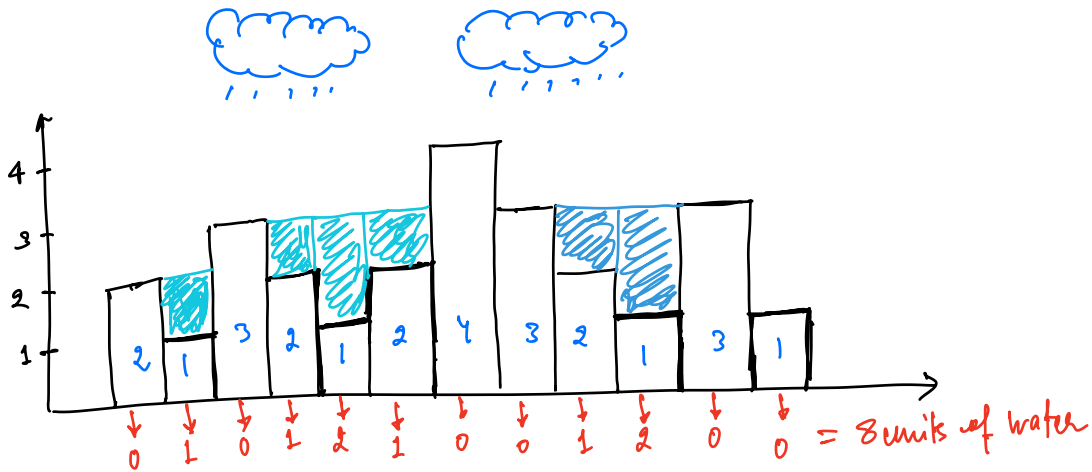
Q1. Rain Water Trapping (Standard interview problem)

Please do not spoil for others if you already know the solution.

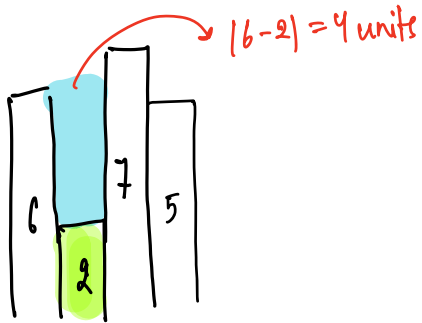
Given N array elements where $arr[i]$ represents height of the building.

Return amount of water trapped by building?

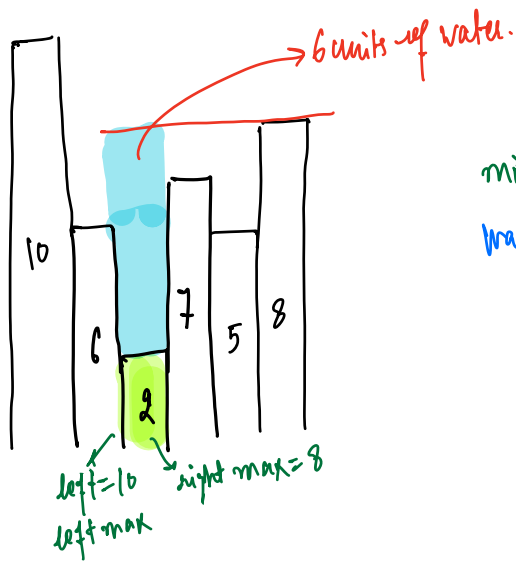
e.g. $arr[] = \{2, 1, 3, 2, 1, 2, 4, 3, 2, 1, 3, 1\}$



e.g.

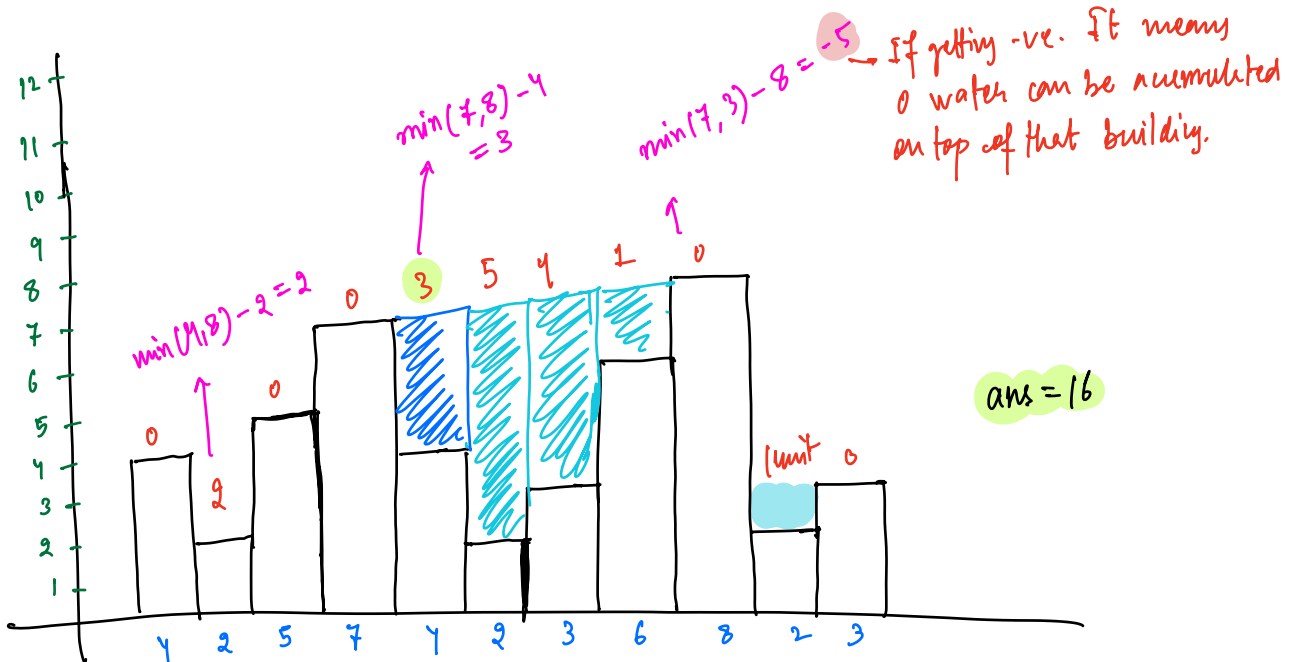


e.g.



$$\min(10, 8)$$

$$\text{water accumulated} = \min(10, 8) - 2 = 8 - 2 = 6 \text{ units}$$



If getting -ve. It means 0 water can be accumulated on top of that building.

for (i=1; i < N-1; i++) {

// for every ith building

lmax = max(0, i-1)

rmax = max(i+1, N-1)

h = min(lmax, rmax)

W = max(h - arr[i], 0);

sum = sum + W;

}

Optimize using pfMax[].

pfMax[i] = max of [0, i]

sfMax[i] = max of [i, N-1]

pfMax[i-1]

Handled.

sfMax[i+1]

O(1)

T.C = O(N + N + N) = O(N)

S.C = O(N + N) = O(N)

O(1)

T.C = O(N * [N + N])

max left

max right

= O(N²)

S.C = O(1)

arr[] = 4, 2, 5, 7, 4, 2, 3, 6, 8, 2, 3

pfMax[] = 4, 4, 5, 7, 7, 7, 7, 7, 8, 8, 8

sfMax[] = 8, 8, 8, 8, 8, 8, 8, 8, 8, 3, 3

Q Given N array elements. find first +ve missing integer.

e.g. 1 $arr[5] = 3 \ -2 \ 1 \ 2 \ 7 \rightarrow 4$

e.g. 2 $arr[7] = -9 \ 2 \ 6 \ 4 \ -8 \ 1 \ 3 \rightarrow 5$

Quiz 1 :- Sum of N natural numbers $\rightarrow$

Quiz 2 :- Given $arr[8] = \{2, 4, -1, -6, 3, 7, 8, 4, -3\}$, find first missing +ve integer $\rightarrow 1$

Quiz 3 Given $arr[6] = \{1, 0, -5, -6, 4, 2\}$ find first missing +ve integer $\rightarrow 3$

Quiz 4 $\{1, 2, 5, 6, 4, 3\} \rightarrow 7$

Observations.

1) min. possible answer = 1

2) $arr[N]$, what can be the maximum answer? $\rightarrow N+1$

$arr[5] = \{1, 3, 4, 5, 2\} \rightarrow \{6\}$
 $\downarrow$
 N $(N+1)$

Ans can only lie in the range $\rightarrow [1, N+1]$ $\rightarrow$ observation

Solution:-

```
for (ele = 1; ele <= (N+1); ele++) {  
    if ele is not present in arr[] → iterate on array and check.  
    // ele is your ans.  
    return ele;  
}
```

$$T.C = O(N^2)$$

$$S.C = O(1)$$

```
for (ele = 1; ele <= N ; ele++) {  
    if ele is not present in arr[] → iterate on array and check.  
    // ele is your ans.  
    return ele;  
}  
return N+1;
```

$$T.C = O(N^2)$$

$$S.C = O(1)$$

Solution:-

HashSet <int> hs;

→ insert all elements in hs (Please do this).

```
for (ele = 1; ele <= N; ele++) {  
    // if element is not present in hs  
    return ele;  
}  
return N+1;
```

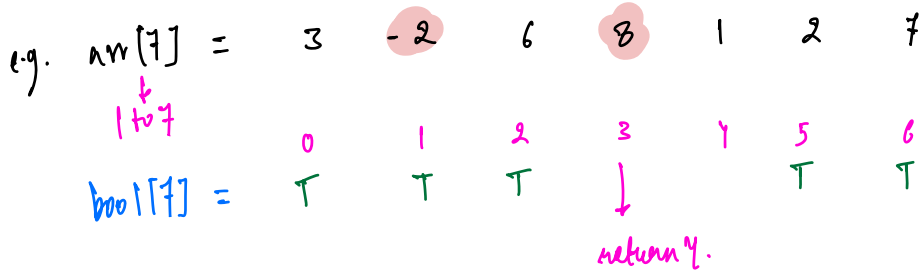
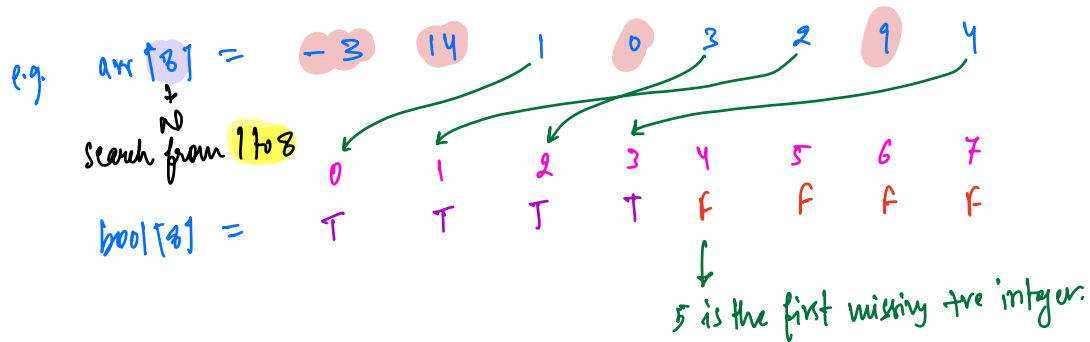
$$T.C = O(N+N) = O(N)$$

$$S.C = O(N)$$

Condition 1 :- You cannot use any in-built functions.

sum of all the elements $-\frac{N(N+1)}{2}$ X

We can take a boolean array of size N.



Pseudo Code:-

bool ch[N];

for (i=0; i < N; i++) {

 ele = arr[i];

 if (ele >= 1 && ele <= N) {

 ch[ele-1] = True;

 }

for (i=0; i < N; i++) {

 if (ch[i] == false) {

 return i+1;

 }

return N+1;

Condition 2 :- No extra space / No libraries.

eg. arr[5] = 5 4 1 -2 3
 ↓
 [1-5]

arr[5] = 0 1 2 3 4
 -5 4 1 -2 -3
 ↓
 +2

Replace all elements <= 0 with +7

eg. arr[5] = 0 1 2 3 4
 5 4 1 7 3
 ↓ ↓ ↓ ↓ ↓
 [1-5] -5 -1 -7 -3

Ans = 2

Pseudo Code:-

- 1) Iterate on array, $arr[i] \leq 0$, replace with $N+1$
- 2) Modify the arr array.

```
for (i=0; i < N; i++) {  
    ele = abs(arr[i]);  
    if (ele >= 1 & ele <= N) {  
        ind = ele - 1;  
        arr[ind] = -1 * abs(arr[ind]);  
    }  
}
```

- 3) Iterate array and get your first missing +ve integer.

```
for (i=0; i < N; i++) {  
    if (arr[i] > 0) {  
        return i+1;  
    }  
}  
return N+1;
```

eg. $arr[7] =$

	0	1	2	3	4	5	6
$arr[i]$	-2	1	3	2	6	3	8
$N=7$							
	-9	-1	-3	2	6	-3	8

↓
return 4

$ans = 4.$